

Senior AI Engineer Take-Home Task – Multi-Source Document Retrieval

As a senior AI engineer at PwC, you're building a prototype for a problem that comes up constantly: PwC manages large collections of client documents — spreadsheets full of product specifications and Word files packed with regulatory guidance — across multiple countries and clients. The catch is that these documents blend two very different kinds of information. Some of it is structured — concrete values, thresholds, and limits that belong in a table (e.g., "Lead content must not exceed 0.05% in the EU"). The rest is unstructured — narrative regulations, usage policies, and compliance guidance (e.g., "Additional ventilation is recommended in hospitals and elderly care homes").

Users like compliance officers, product managers, and auditors need to ask plain-language questions and get accurate, sourced answers — without accidentally seeing another client's data.

You have two clients to work with:

Aurora Paints manufactures the EcoSafe line — environmentally-conscious paints for consumer and commercial use. Their data covers chemical composition limits (VOC caps, lead thresholds), eco-certification parameters, and regional compliance rules. You're given `aurora_product_data.xlsx` (hundreds of spec rows across products and regions) and `aurora_product_brief.docx` (regulatory narrative, inline spec entries in bullet/text form, and an embedded image containing key spec details).

Horizon Coatings produces the UltraSafe line — performance coatings engineered for safety in demanding environments like hospitals, industrial facilities, and elderly care homes. Their data emphasizes safety thresholds, application guidelines, and regulatory considerations for sensitive settings. You're given `horizon_product_data.xlsx` (hundreds of spec rows, different schema from Aurora) and `horizon_product_brief.docx` (same mix: narrative text, inline specs, and an embedded spec image).

Both clients provide logically similar information — product, region, parameter, value, unit — but their spreadsheet schemas differ (different column names and structures). Spec-like data isn't always in tables; some appears as bullets or inline text in the Word docs. Each Word doc also contains at least one spec entry matching its client's spreadsheet schema. You can handle missing data values if any, missing columns can be null/None.

Your prototype must:

1. Ingest all four files (two spreadsheets, two Word docs).
2. Separate and store information into a structured store (for specification data) and an unstructured store (for narrative/regulatory text).

3. Answer natural-language questions by automatically routing to the right store(s), applying appropriate retrieval, providing explainable answers (showing source information), and enforcing strict client isolation — Aurora's data never leaks into Horizon's answers and vice versa.
4. Support conversational memory with user sessions — continuous chat, not one-shot Q&A.

Note on client isolation: In production, the client ID would come from the session. For this task, pass the client name in the query and access only that client's data.

Example questions the system should handle:

- *"What is the maximum allowed VOC content for this product in the EU?"*
- *"Which client has stricter VOC limits for interior wall paint in the EU?"*
- *"Summarise the key regulatory considerations for using this paint in sensitive environments."*
- *"As Client Aurora Paints, what are the applicable limits for my products without mixing in any other client's data?"*

Constraints & Freedom

- **You choose** all technical details: libraries, data models, databases, LLMs, retrieval strategies.
- **You can use free LLM APIs** : Gemini, Groq, OpenRouter.
- **You have N calendar days** from receiving this task.

4. Functional Requirements

4.1 Ingestion and Information Separation

Your solution must:

- 1) Read the spreadsheet and Word files for both clients.
- 2) Extract content from these files.
- 3) Decide, based on your own logic, which parts of the content are:
 - Specification-like information that belongs in a structured store.
 - Policy / regulatory / narrative information that belongs in an unstructured store.
 - Client-related information that allows you to keep data for different clients separated.
- 4) Populate both stores accordingly so that they can later be used to answer questions.

Note on schemas:

How you detect and separate structured vs. unstructured information is up to you. You should be prepared to explain the heuristics, rules, or models you used, and their limitations.

4.2 Retrieval, Routing and Advanced Technique

Your solution must allow a user to submit a natural-language query and must:

1) Decide which information source(s) to use for each query:

- Only the structured store.
- Only the unstructured store.
- Both, combined in a meaningful way.

This “router” can be implemented however you see fit (rules, models, or a combination), but you must be able to explain your design and trade-offs.

2) Retrieve relevant information from the chosen store(s) and generate a final answer.

4.3 User Interaction and Queries to Demonstrate

You are not required to build a graphical user interface. A notebook, REPL or simple script is sufficient.

Please demonstrate at least the following five queries and briefly analyse the behaviour of your system. and be prepared to demonstrate them live during the task discussion meeting along with additional examples.

Query 1

“For client Aurora Paints, what is the maximum zinc content allowed in the finished EcoSafe Interior Wall Paint for the EU?”

Query 2

“For client Aurora Paints, considering **EcoSafe Ceiling Paint**, **EcoSafe Exterior Facade** and **EcoShield Floor Coating** in the **EU**, what is the **maximum internal VOC limit** in g/L across these products? ”

Query 3

“According to the guidance for client Horizon Coatings’ UltraSafe Interior Wall Paint, in which types of rooms is enhanced ventilation or longer airing-out periods specifically recommended?”

Query 4

“Comparing client Aurora Paints and client Horizon Coatings, which client sets a stricter VOC limit for interior wall paint in the EU, and what additional usage guidance is mentioned for sensitive environments across their products?”

Query 5

For client Aurora Paints, what internal VOC limit in g/L is set for EcoSafe Kitchen & Bath in the EU for typical residential projects?

For each of these queries in the video, you should:

- Show the query text.
- Show the final answer.
- Briefly explain which store(s) were used and why that makes sense.
- Comment on whether the behaviour matches your expectations and, if not, why.

4.4 Bonus Objectives (Optional)

These are not required but will be considered a plus:

- Streaming answers: stream answers incrementally instead of only returning the final string. Briefly describe your approach in the README if implemented.
- Multilingual support (for example Arabic): describe how you would extend your design to handle additional languages and any special considerations.
- Image-aware behavior: make use of the images in the word files, try to query on their content and analyze the results.

5 Deliverables

5.1 Recorded Video (Maximum 10 Minutes)

Submit a screen-recorded video of at most 10 minutes that covers:

- Walkthrough of the implementation: Text Extraction, ingestion, separation into structured vs. unstructured stores, client separation, Retrieval, query routing, and the answers to the 5 queries with your interpretation in these answers.
- High-level design: how your system addresses the business need of answering compliance and specification questions over mixed documents for multiple clients, and a short architecture overview.

5.2 Task Code and any related files, diagrams ...